

11. Let  $L_1 = \{aaa\}^*$ ,  $L_2 = \{a, b\}\{a, b\}\{a, b\}\{a, b\}$ , and  $L_3 = L_2^*$ . Describe the strings that are in the languages  $L_2$ ,  $L_3$ , and  $L_1 \cap L_3$ .

For Exercises 12 through 37, give a regular expression that represents the described set.

12. The set of strings over  $\{a, b, c\}$  in which all the  $a$ 's precede the  $b$ 's, which in turn precede the  $c$ 's. It is possible that there are no  $a$ 's,  $b$ 's, or  $c$ 's.
13. The same set as Exercise 12 without the null string.
14. The set of strings of length two or more over  $\{a, b\}$  in which all the  $a$ 's precede the  $b$ 's.
15. The set of strings over  $\{a, b\}$  that contain the substring  $aa$  and the substring  $bb$ .
16. The set of strings over  $\{a, b\}$  in which the substring  $aa$  occurs at least twice. *Hint:* Beware of the substring  $aaa$ .
17. The set of strings over  $\{a, b, c\}$  that do not contain the substring  $aa$ .
18. The set of strings over  $\{a, b\}$  that do not begin with the substring  $aaa$ .
19. The set of strings over  $\{a, b\}$  that do not contain the substring  $aaa$ .
20. The set of strings over  $\{a, b\}$  that do not contain the substring  $aba$ .
21. The set of strings over  $\{a, b\}$  in which the substring  $aa$  occurs exactly once.
22. The set of strings over  $\{a, b, c\}$  that begin with  $a$ , contain exactly two  $b$ 's, and end with  $cc$ .
23. The set of strings over  $\{a, b\}$  that contain the substring  $ab$  and the substring  $ba$ .
24. The set of strings over  $\{a, b, c\}$  that contain the substrings  $aa$ ,  $bb$ , and  $cc$ .
25. The set of strings over  $\{a, b, c\}$  in which every  $b$  is immediately followed by at least one  $c$ .
26. The set of strings over  $\{a, b, c\}$  with length three.
27. The set of strings over  $\{a, b, c\}$  with length less than three.
28. The set of strings over  $\{a, b, c\}$  with length greater than three.
29. The set of strings over  $\{a, b\}$  in which the number of  $a$ 's is divisible by three.
30. The set of strings over  $\{a, b, c\}$  in which the total number of  $b$ 's and  $c$ 's is three.
31. The set of strings over  $\{a, b\}$  in which every  $a$  is either immediately preceded or immediately followed by  $b$ , for example,  $baab$ ,  $aba$ , and  $b$ .
32. The set of strings of odd length over  $\{a, b\}$  that contain the substring  $bb$ .
33. The set of strings of even length over  $\{a, b, c\}$  that contain exactly one  $a$ .
34. The set of strings over  $\{a, b, c\}$  with an odd number of occurrences of the substring  $ab$ .
35. The set of strings of odd length over  $\{a, b\}$  that contain exactly two  $b$ 's.
36. The set of strings over  $\{a, b\}$  with an even number of  $a$ 's or an odd number of  $b$ 's.
37. The set of strings over  $\{a, b\}$  in which the number of  $a$ 's is even and the number of  $b$ 's is odd. This is tricky; a string of length zero is not in the set.
38. Use the regular expressions  $a^+b^+c^+$  and  $a^+b^+c^+a^+b^+c^+$  to describe the set of strings over  $\{a, b, c\}$  in which the substrings  $abc$  and  $abcabc$  occur.
  - a)  $(ba)^+(a^*b^* \cup a^*b^*c^*a^*b^*c^*)^*$
  - b)  $b^+(a^*b^* \cup \lambda)b$
  - c)  $(a \cup b)^* = (a^*b^*)^*$
  - d)  $(a \cup b)^* = (a^*b^*)^*$
  - e)  $(a \cup b)^* = (b^*a^*)^*$

### Bibliographic Notes

Regular expressions were first used in the context of computer networks. McNaughton [1960] showed that the operations of intersection and union on regular expressions can be expressed using regular expressions.